FAILURE OF THE PROPOSED HAMILTONIAN NEPV RAYLEIGH IDENTITY

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ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: The proposed NEPv matrix does not reproduce the product-state Rayleigh quotient when the Hamiltonian sum includes a same-site term with $i = j$. *Certificate:* The certificate consists of exact 2×2 integer matrices and the integer vector $(1, 2)^T$, for which the two assertedly equal values are 0 and $-4/25$.

Posed by Amsel et al., §6.2 (2026); counterexample found by OpenAI Codex (2026-08); formalized by OpenAI Codex; audited by OpenAI Codex; contributed by Suvrit Sra.

Section 6.2 considers $n, d \in \mathbb{N}$, nonzero vectors $x_1, \ldots, x_d \in \mathbb{C}^n$, and matrices $F_i, G_{i,j}, K_{i,j} \in \mathbb{C}^{n \times n}$ that are Hermitian and have norm at most one. It defines

$$H = \sum_{i=1}^{d} \left(F_i^{\langle i \rangle} + \sum_{j=1}^{d} G_{i,j}^{\langle i \rangle} K_{i,j}^{\langle j \rangle} \right)$$

and the product-state Rayleigh quotient

$$f(x_1, \ldots, x_d) = \frac{(x_1 \otimes \cdots \otimes x_d)^H H (x_1 \otimes \cdots \otimes x_d)}{\prod_{j=1}^{d} x_j^H x_j}.$$

The proposed block diagonal matrix A has blocks

$$A_i = \left(\frac{\sum_{j=1}^{d} x_j^H x_j}{x_i^H x_i} \right) \left(F_i + \sum_{j=1}^{d} G_{i,j} \frac{x_j^H K_{i,j} x_j}{x_j^H x_j} \right).$$

The displayed source statement writes the stack through x_n, although A has d blocks and the surrounding text declares the NEPv dimension to be nd. The only dimensionally consistent reading is the stack through x_d; the counterexample below uses $d = 1$, so that $z = x_1$.

Problem 1 (As posed: The assertion appears immediately after equation (20) in Section 6.2 of the Simons workshop report (Amsel et al., 2026). That section is scribed by Edgar Solomonik and records discussion with Elias Jarlebring, Florian Schafer, Maryam Dehghan, Tamara Kolda, and Chris Camaño; it says the question is based on a pre-workshop question of Elias Jarlebring.)**.** Then, we have that with

$$z = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \frac{z^H A(z) z}{z^H z} = f(x_1, \ldots, x_d).$$

Date: 9 August 2026.

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

The construction in Section 6.2 of (Amsel et al., 2026) replaces a same-site expectation of a product by the product of two expectations. Those quantities need not agree.

Theorem 2 ([**Disproof**]: proposed NEPv Rayleigh identity)**.** *The displayed Rayleigh-quotient identity following equation (20) is false for the stated matrix formulas. This refutes an assertion in the setup of Problem 6.2, not the problem's request for an efficient algorithm.*

Proof. Set $n = 2$ and $d = 1$, and take

$$F_1 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad G_{1,1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad K_{1,1} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}, \qquad x_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

All three matrices are Hermitian and have norm at most one. Moreover, $G_{1,1}$ and $K_{1,1}$ commute, so the matrix H below is Hermitian. Since $d = 1$, equation (18) gives

$$H = G_{1,1}K_{1,1} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore the objective in equation (19) is

$$f(x_1) = \frac{x_1^H H x_1}{x_1^H x_1} = \frac{0}{5} = 0.$$

On the other hand, the scalar prefactor in equation (20) is one and

$$\frac{x_1^H K_{1,1} x_1}{x_1^H x_1} = -\frac{4}{5}, \qquad A(x_1) = -\frac{4}{5} G_{1,1}.$$

It follows that

$$\frac{x_1^H A(x_1) x_1}{x_1^H x_1} = -\frac{4}{5}\frac{1}{5} = -\frac{4}{25} \neq 0 = f(x_1).$$

The failed step is specific and algebraic: for $i = j$, equation (18) contains

$$\frac{x_i^H G_{i,i} K_{i,i} x_i}{x_i^H x_i},$$

whereas equation (20) produces

$$\left(\frac{x_i^H G_{i,i} x_i}{x_i^H x_i}\right)\left(\frac{x_i^H K_{i,i} x_i}{x_i^H x_i}\right).$$

Hermiticity of the two factors does not make these expressions equal; the counterexample uses commuting factors and even has $H = 0$. □

Code and certificate

The case is the directory `counterexamples/hamiltonian-nepv-identity/` of the archive, at commit `3dcaa0f`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/hamiltonian-nepv-identity
python verify.py
```

- Theorem 2 (result `hamiltonian-nepv-identity`): exact arithmetic.
- `artifacts/certificate.json`: Exact rational Hermitian matrices and vector, with the two unequal Rayleigh quotients

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by OpenAI Codex (2026-08). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

ACKNOWLEDGEMENTS

The author (SS) thanks the Alexander von Humboldt (AvH) Foundation for their generous support via a Humboldt Professorship in AI.

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